

ADITYA RANJAN SWAIN

Full-Stack Developer | Front-End Engineer | MERN Stack | React.js | Node.js

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PROFESSIONAL SUMMARY

Final-year B.Tech Computer Science student at Trident Academy of Technology (CGPA: 8.0/10, graduating May 2026) with hands-on experience building full-stack web applications using the MERN stack and Python/Flask. I have independently designed and shipped three production-grade projects covering real-time collaboration tools, role-based authentication systems, and AI-powered web platforms. Comfortable working across the full stack — from crafting responsive React frontends and RESTful APIs in Node.js/Express, to managing MongoDB and MySQL databases. I write clean, maintainable code and enjoy solving real problems through thoughtful UI and solid backend architecture. Looking for a Full-Stack or Front-End Developer role where I can contribute from day one.

TECHNICAL SKILLS

Frontend: React.js, JavaScript (ES6+), HTML5, CSS3, Bootstrap, Responsive Design, Redux Toolkit

Backend: Node.js, Express.js, Python, Flask, REST API Development, Socket.io, Flask-WTF

Databases: MongoDB (including Geospatial Indexing), MySQL, Firebase, Mongoose ODM

AI / ML Integration: DeepFace, OpenCV, Dlib (Shape Predictor 68 Face Landmarks), Docker, Docker Compose

Tools & Platforms: Git, GitHub, VS Code, Postman, Stripe API, Firebase Auth, Google Maps API, Leaflet.js

Concepts: Full-Stack Web Development, MVC Architecture, Role-Based Access Control (RBAC), Real-Time Communication (WebSockets), Optimistic UI Updates, Agile Development, Responsive UI Design, Third-Party API Integration

PROJECTS

Smart AI-Based Proctoring System | *Python · Flask · MySQL · DeepFace · OpenCV · Dlib · Stripe · Docker*

- Built a full-stack AI-powered exam proctoring platform from scratch with role-based access for Professors and Students, handling the entire system — from user authentication and exam creation through to live monitoring and payment integration.
- Implemented biometric authentication using DeepFace, verifying student identity against registered facial profiles in real time before allowing exam access, making the system resistant to impersonation.
- Developed a live AI proctoring engine using OpenCV and Dlib's 68-point facial landmark model to continuously track face position, detect gaze direction, flag multiple faces, and log suspicious behaviour automatically throughout each session.
- Supported three distinct exam types — MCQ (with CSV upload, negative marking, and calculator), subjective long-form answers, and practical coding exams with built-in code compilation for 15+ languages including C, C++, Java, Python, and Node.js.
- Integrated Stripe API for secure exam credit purchases, allowing Professors to buy hosting credits and manage their exam quota through a clean billing interface.
- Containerised the full application using Docker and Docker Compose for consistent, reproducible deployment across environments, with MySQL as the primary relational database and Flask-Session for secure session management.

Collaborative Enterprise Project Management Workspace | *React.js · Node.js · Express.js · MongoDB · Socket.io · Redux Toolkit*

- Designed and built a multi-user real-time project management platform similar to Trello, featuring drag-and-drop kanban boards, workspace dashboards, and live state synchronisation across all connected users simultaneously.
- Implemented real-time drag-and-drop using @hello-pangea/dnd on the frontend, broadcasting task position updates instantly to every active user via Socket.io — so when one user moves a card, all others see it move without any page refresh.

- Built optimistic UI updates so the interface responds immediately to user actions before the server confirms the database write. If the API call fails, the card smoothly rolls back to its original position, keeping the experience fast and reliable.
- Developed a background audit trail system that logs every workspace action — who moved what task, from which column to which, and at what time — stored in a structured MongoDB schema for historical review and accountability.
- Implemented role-based workspace permissions using Express middleware and JWT authentication, giving workspace owners control over member roles and access levels across multiple independent workspaces.

Advanced Remote Examination & Assessment Portal | *React.js · Redux Toolkit · Node.js · Express.js · MongoDB · Chart.js*

- Built a full-stack online examination platform with three separate role-based portals — Admin, Teacher, and Student — each with a purpose-built React dashboard and strictly scoped API access enforced on the Express backend.
- Developed a rich exam creation interface for Teachers using a React rich text editor, supporting dynamic question banks, multiple-choice options, automated grading logic, and configurable time limits per exam.
- Implemented automated proctoring logs on the frontend that detect and timestamp browser blur events, tab switches, and focus changes during active exams, saving a structured log array to MongoDB for Teacher review after submission.
- Designed a complex MongoDB schema linking exam papers, individual questions, per-student answer submissions, and automated grade calculations — supporting fast querying and accurate result generation at scale.
- Built an analytics dashboard for Teachers using Chart.js to visualise class performance, pass/fail ratios, score distributions, and average exam durations — giving instructors clear, actionable insights without manual data processing.

EDUCATION

B.Tech in Computer Science Engineering

2022 – May 2026

Trident Academy of Technology, Bhubaneswar | CGPA: 8.0 / 10 | Relevant: Data Structures & Algorithms, DBMS, Web Technologies, Operating Systems

Class XII — Computer Science (CBSE)

2022

Sainik School Bhubaneswar | 65%

Class X — Mathematics (CBSE)

2020

Sainik School Bhubaneswar | 80%

CERTIFICATIONS

Full-Stack Web Development (MERN Stack) — Self-directed / GitHub Projects

2024 – 2025

AI/ML Training Program — CTTC MSME, Bhubaneswar

2025

ACHIEVEMENTS & ACTIVITIES

- Built and deployed three production-grade full-stack web applications independently while completing a full-time B.Tech degree, all available on GitHub with documentation.
- Developed a real-world AI proctoring system integrating biometric verification, live video analysis, and Stripe payments — a level of feature depth rarely seen in student projects.
- Active GitHub contributor with multiple repositories covering full-stack and AI-integrated web development projects.
- Maintained a CGPA of 8.0/10 throughout the B.Tech programme at Trident Academy of Technology, Bhubaneswar.